C Y B E R T H R E A T I N T E L L I G E N C E R E P O R T

Dust Specter

State-Sponsored OSINT Operation Targeting Iraqi Diplomatic Infrastructure

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1. Executive Summary

Dust Specter is a state-sponsored cyber intelligence operation first detected in July 2025 and active through January 2026. The operation targeted senior officials within Iraq's Ministry of Foreign Affairs and diplomatic apparatus through two distinct attack chains, designated SPLITDROP/TWINTASK/TWINTALK (Chain 1) and GHOSTFORM (Chain 2).

This report argues that Dust Specter is not a conventional cyberattack campaign. It is an OSINT-based counter-intelligence operation executed by a state-level intelligence unit, designed to monitor Iraq's diplomatic traffic and detect early signals of Iraq's geopolitical realignment away from Iran. The operation's timeline, targeting profile, and operational posture are consistent with an actor in crisis mode following the Twelve-Day War of June 2025, in which Iran sustained severe military and intelligence losses.

The operation's infrastructure was withdrawn in January 2026, two months before public disclosure by Zscaler in March 2026. This withdrawal preceded the February 28, 2026 US-Israeli strike on Iran, in which Iran's senior intelligence leadership was killed in the opening wave. The convergence of these events suggests that Dust Specter's intelligence collection failed to prevent Iran's strategic surprise, either because the operation was compromised before critical intelligence could be acted upon, or because the intelligence gathered was insufficient to anticipate the scale of the threat.

ASSESSMENT	DETAIL
Threat Actor	Unknown Iran attribution assessed with LOW-MODERATE confidence. False flag scenario remains active.
Operation Type	State-sponsored OSINT/counter-intelligence operation
Target	Iraqi Ministry of Foreign Affairs, diplomatic officials
Active Period	July 2025 - January 2026 (7 months)
Infrastructure Status	Withdrawn prior to public disclosure operator-initiated
Operation Outcome	Assessed FAILED Iran sustained strategic surprise on Feb 28, 2026

2. Methodology

This report is produced entirely from open source intelligence (OSINT). No classified, restricted, or non-public sources were accessed. Primary technical data is derived from Zscaler ThreatLabz's March 2026 disclosure. Geopolitical context is derived from open reporting by Reuters, AP, and regional news sources.

The analytical process applied in this report combines standard CTI methodology with a personal analytical framework developed by the author from prior operational experience in human intelligence environments. This framework is not publicly documented, not derived from open source literature, and is disclosed here solely in the interest of methodological transparency.

Personal Analytical Framework

The framework consists of four sequential modules applied in order as the analysis develops. Each module serves a distinct function and feeds into the next.

CODE	MODULE	FUNCTION IN THIS ANALYSIS
T.E	Detection	Applied to identify retention patterns and concealed intent within Dust Specter's operational profile. Used to flag anomalies in what was present in the technical artifacts and what was absent from public geopolitical discourse surrounding this campaign.
FM	Scenario Mapping	Applied to construct multiple independent scenario hypotheses before any conclusion was drawn. Each scenario was built as a standalone construct to prevent early convergence on a preferred explanation. Four scenarios were constructed; one survived the full analytical process.
NM	Anti-Bias Check	Applied before each major analytical conclusion. Used to stress-test assumptions by examining the same evidence from the perspective of an actor with opposing interests, and to evaluate the cost of being wrong in both directions before committing to an assessment.
RH	Consequence Mapping	Applied to assess the scale, intensity, persistence, and reversibility of each scenario's consequence profile. Used to determine which scenario carried the most operationally coherent outcome given the full body of observed evidence.

The analysis began with four scenario constructs. Three were deprioritized as the evidence base developed. The surviving scenario and its supporting evidence are presented in Section 4.

Confidence Level Scale

LEVEL	DEFINITION
HIGH	Supported by direct technical evidence or corroborated open source reporting
MODERATE	Supported by circumstantial evidence or pattern correlation
LOW	Inferred from limited indicators requires additional evidence to confirm

3. Technical Analysis

3.1 Attack Chain Overview

Dust Specter employs two structurally distinct attack chains deployed against the same target set. The existence of dual mechanisms is assessed as operationally significant and is discussed in Section 3.4.

COMPONENT	CHAIN	FUNCTION
SPLITDROP	Chain 1	Initial dropper. Deploys TWINTASK and TWINTALK to disk.
TWINTASK	Chain 1	Worker module. Polls local file every 15 seconds for Base64-encoded PowerShell commands.
TWINTALK	Chain 1	C2 orchestrator. Beacons to remote server, writes commands to local file for TWINTASK consumption. Separates network and execution roles.
GHOSTFORM	Chain 2	Consolidated single binary. In-memory PowerShell execution with minimal filesystem footprint. Embeds Google Form masquerading as Iraqi MFA survey.

3.2 C2 Infrastructure

- Primary domain: meetingapp[.]site masquerades as Cisco Webex for Government
- Beaconing interval: randomized 108-180 seconds to evade pattern-based network detection
- URI path construction: dynamic, with appended checksum to verify requests originate from genuinely infected systems
- Geofencing: active requests from outside target geography receive no payload
- Domain ca[.]iq: compromised legitimate Iraqi government domain used for GHOSTFORM payload hosting

The use of ca[.]iq, a domain belonging to Iraq's own government infrastructure, is assessed as confirmation of insider access within Iraqi government systems.

CONFIDENCE: HIGH

3.3 GHOSTFORM — Active Data Collection Vector

GHOSTFORM embeds hardcoded Google Form URLs within its binary. These forms, written in Arabic, are automatically opened in the victim's browser upon execution. The forms impersonate official surveys from Iraq's Ministry of Foreign Affairs.

This mechanism represents a dual-layer collection approach GHOSTFORM exfiltrates system data and command execution output through its C2 channel, while simultaneously soliciting voluntary disclosure of information from the victim through a socially engineered form. The

combination of passive collection and active solicitation is consistent with an intelligence operation focused on understanding the victim's knowledge and access, not merely their system.

3.4 Dual Mechanism — Operational Significance

The existence of two structurally independent attack chains is not interpreted as technical redundancy. Redundancy in malware deployment is typically implemented at the component level, not the architectural level. Two fully separate chains against the same target set reflects operational anxiety, not engineering preference.

The interpretation applied in this report is that the dual mechanism functions as a continuity guarantee: if Chain 1 is detected and neutralized, Chain 2 continues operating independently. This design implies that the operator assessed a meaningful probability of partial detection and built against it.

This design posture is consistent with an operator who has insider knowledge of the target environment and therefore confidence that both chains can be deployed but who cannot guarantee that either will remain undetected indefinitely. The dual mechanism is the architectural expression of that uncertainty.

CONFIDENCE: MODERATE

3.5 AI-Assisted Development — Artifact Analysis

Zscaler identified indicators of AI assisted code generation within the TWINTALK binary. Two specific artifacts were noted:

- Emoji characters used as functional return values within the set_in_measure function: the warning emoji as a null-data indicator, and a speech bubble emoji as a data prefix for payloads exceeding 900,000 characters
- Hardcoded seed value 0xABCDEF within the checksum generation function a placeholder characteristic of AI-generated code that was not replaced before deployment

These artifacts are not operationally functional beyond their immediate role in the code. No evidence suggests they were inserted as intentional signals or covert markers. They are assessed as residual artifacts from an accelerated development process in which AI tooling was used to produce functional malware code under time pressure.

The presence of these artifacts is consistent with the operational timeline: Dust Specter's first detected activity was July 2025, immediately following Iran's severe military losses in the Twelve-Day War of June 2025. An intelligence unit under pressure to deploy rapidly would accept minor code hygiene tradeoffs in exchange for operational speed.

The specific model or platform used for code generation cannot be determined from available artifacts. The emoji pattern is consistent with modern instruction-tuned language models but is not diagnostic of any specific system. Attribution of the AI tooling is assessed as LOW confidence.

CONFIDENCE: MODERATE — AI usage confirmed. Platform unknown.

3.6 Infrastructure Withdrawal

Zscaler's analysis notes that the GHOSTFORM payload URL at meetingapp[.]site/webexdownload was inaccessible at the time of analysis in January 2026. The infrastructure was withdrawn prior to Zscaler's public disclosure in March 2026 a gap of approximately two months.

This withdrawal is assessed as operator-initiated rather than externally forced. The operational security profile of Dust Specter geofencing, dynamic checksum URIs, randomized beaconing intervals, separation of network and execution roles is consistent with an operator who actively monitors for detection signals and is capable of a disciplined withdrawal when those signals appear.

A forced takedown by a hosting provider or law enforcement would not explain the timing differential between withdrawal and public disclosure. An operator-initiated withdrawal does.

CONFIDENCE: HIGH

4. Strategic Assessment

This section constitutes a strategic assessment and is clearly separated from the technical findings above. All assessments in this section carry explicitly stated confidence levels. Readers should treat this section as analytical judgment, not confirmed fact.

4.1 Iraq as Early Warning Post

The targeting profile of Dust Specter — senior diplomatic officials with access to cross-ministry coordination and foreign policy communication is not consistent with financial or espionage objectives oriented toward Iraq itself. The data accessible through these targets is most valuable to an actor seeking to understand Iraq's diplomatic relationships with other regional actors: who Iraq is talking to, how often, and whether those conversations are changing.

Iraq occupies a structurally unique position in the regional landscape. It shares a border with Iran, maintains economic dependencies on Iranian energy infrastructure, and simultaneously hosts significant US military presence and draws investment from Gulf states aligned with the US. Any meaningful shift in Iraq's diplomatic alignment would be detectable first through changes in its foreign ministry's communication patterns before it would become visible through public policy, treaty negotiations, or economic agreements.

This report assesses that Dust Specter's primary intelligence objective was not to collect information about Iraq, but to use Iraq as a monitoring station for the broader regional alignment landscape. Iraq's diplomatic traffic is a leading indicator of regional stability or instability.

CONFIDENCE: MODERATE

4.2 Operational Context — Iran Post-June 2025

The Twelve-Day War of June 2025 left Iran's military and intelligence infrastructure severely degraded. Key facilities were struck, senior military leadership was killed, and Iran's air defense capabilities were substantially reduced. This context is directly relevant to understanding why Dust Specter began in July 2025.

An intelligence service that has lost significant human and technical assets does not pause collection. It accelerates it, under conditions of reduced capability and heightened urgency. The AI-assisted development artifacts in Dust Specter's code are consistent with exactly this dynamic: rapid deployment under time pressure, with acceptable tradeoffs in code hygiene.

By late 2025, Iran also faced significant internal pressure. Reporting from November-December 2025 describes substantial domestic unrest, with protests that began as economic grievances expanding into broader demands. A government under simultaneous external military pressure and internal political pressure has a compounded need for foreign intelligence, and a compounded incentive to accept operational risk in collecting it.

CONFIDENCE: MODERATE

4.3 Analytical Convergence

The analytical process described in Section 2 began with four scenario constructs. As the technical artifacts, target profile, geopolitical context, and timeline were integrated, three scenarios became progressively unsupported by the evidence. The analysis converged on a

single coherent explanation without forcing closure on the alternatives they were deprioritized because the evidence did not sustain them, not because they were excluded by assumption.

The surviving assessment is this Dust Specter is a proactive defensive intelligence operation. The operator was not extracting value from Iraq. The operator was watching Iraq to detect whether the regional equilibrium was breaking. Iraq was chosen because it is the first node in the regional network that would show a realignment signal before that signal became visible anywhere else.

Attribution to a state intelligence unit remains at LOW-MODERATE confidence. The false flag possibility has not been resolved. This is acknowledged in the Limitations section and does not change the operational assessment of what Dust Specter was built to do.

CONFIDENCE: MODERATE

4.4 Operation Outcome Assessment

On February 28, 2026, the United States and Israel conducted coordinated strikes against Iran, killing the Supreme Leader and senior military officials. In the opening wave of that strike, multiple senior leaders of Iran's Ministry of Intelligence were killed including the Deputy Minister for Israeli Affairs, the head of the Espionage Division, and the Deputy for Internal Security.

The killing of the intelligence leadership in the first strike indicates that the US and Israel had precise, current knowledge of their locations. This is inconsistent with an Iranian intelligence apparatus that had successfully monitored for the threat. It is consistent with an Iranian intelligence apparatus that had been operationally blinded.

Separately, reporting from November 2025 documents a direct meeting between US Special Envoy Thomas Barrack and Iraqi Prime Minister Al-Sudani in Baghdad, in which Barrack warned of an impending Israeli strike if Iran-aligned militias continued operations. Iraqi forces subsequently shot down Iranian drones targeting US assets near Baghdad Airport during the February 28 strikes.

Dust Specter was designed to detect exactly this kind of shift: Iraq moving from passive coexistence with Iran's regional presence toward active alignment with the opposing axis. The available evidence suggests it failed to do so in time to be actionable.

CONFIDENCE: MODERATE — Based on open source correlation, not direct confirmation.

5. Indicators of Compromise

5.1 Network Indicators

TYPE	INDICATOR	NOTES
meetingapp[.]site	Domain	C2 masquerades as Cisco Webex for Government
lecturegenieltd[.]pro	Domain	Secondary C2 infrastructure
ca[.]iq	Domain	Compromised legitimate Iraqi government domain GHOSTFORM payload hosting
/webexdownload	URI Path	GHOSTFORM payload delivery path on meetingapp[.]site

5.2 File Indicators

TYPE	HASH / NAME
SHA-256	da1775d0fbe99fbc35b6f0b4a3a3cb84da3ca1b2c1bbac0842317f6f804e30a4
SHA-1	848e45088ab83918b2fef09301663cb3759d550e
MD5	38233d77050b5f34aa0c0014d1b8ab3a
Filename	mofa-Network-code.rar

5.3 Behavioral Indicators

- DLL sideloading via legitimate VLC Media Player binary
- File-based polling at 15-second intervals for Base64-encoded PowerShell commands
- Randomized C2 beaconing interval: 108-180 seconds
- Dynamic URI path construction with appended checksum
- In-memory PowerShell execution without disk-based staging (GHOSTFORM)
- Automatic browser launch to Google Forms URL upon execution

6. MITRE ATT&CK Mapping

TECHNIQUE ID	NAME	OBSERVED IN DUST SPECTER
T1566.001	Spearphishing Attachment	RAR archive mofa-Network-code.rar delivered to targeted officials
T1574.002	DLL Side-Loading	SPLITDROP deployed via VLC Media Player sideloading
T1059.001	PowerShell	TWINTASK executes Base64-decoded PowerShell commands from local file
T1027	Obfuscated Files	Base64 encoding of command payloads; in-memory execution in GHOSTFORM
T1071.001	Web Protocols	C2 communication via HTTPS to meetingapp[.]site
T1102	Web Service	Google Forms used as active data solicitation vector
T1547	Boot/Logon Autostart	Persistence mechanism via Chain 1 components
T1041	Exfiltration Over C2	System data and command output exfiltrated via TWINTALK C2 channel
T1195.001	Compromise Software Supply Chain	Compromised ca[.]iq used as trusted delivery infrastructure

7. Analysis Limitations and Transparency Statement

This report is produced by an independent analyst operating exclusively within open source material. The limitations below are not qualifications offered to soften the analysis. They are structural boundaries that any reader applying this report to operational or policy decisions must understand.

Attribution

Attribution to Iran is assessed at LOW-MODERATE confidence. The technical profile, target selection, and geopolitical context are consistent with an Iranian state intelligence operation. They are not exclusive to one. The false flag scenario was constructed and evaluated through the FM module and was not resolved to closure. It remains an open possibility. No reader should treat the attribution in this report as confirmed.

Intelligence Objective

The specific data collected by Dust Specter during its seven-month operational period is unknown. The assessment of the operation's intelligence objective, monitoring Iraq's diplomatic alignment as an early warning indicator, is inferred from target selection, malware capability, and geopolitical context. It is not derived from recovered collection data, because no such data is publicly available.

Geopolitical Assessment

Section 4 constitutes strategic assessment, not technical finding. All conclusions in that section are the analytical judgment of the author, derived from open source reporting and applied through the personal framework described in Section 2. This assessment has not been validated by any intelligence community source, regional expert, or institutional review. Timeline correlations between Dust Specter's activity and geopolitical events are treated as analytically significant but are not causal confirmation.

Scope of OSINT Analysis Against State Operations

Dust Specter is assessed as a state-level intelligence operation. Open source analysis of state operations operates against a fundamental asymmetry: the analyst works from what was disclosed, detected, and published, while the operation was designed specifically to avoid all three. What is visible in the public record is, by definition, what the operator failed to conceal. What was successfully concealed is invisible to this analysis. The conclusions in this report should be understood as the floor of what occurred, not the ceiling.

Confidence levels stated throughout this report exist to help the reader calibrate reliance on each finding. Findings at LOW confidence should inform further investigation. They should not drive decisions.

8. Defensive Recommendations

- Monitor for DLL sideloading activity involving legitimate media player binaries, particularly in environments where VLC or similar software is present
- Implement network monitoring for beaconing behavior with randomized intervals in the 100-200 second range to HTTPS destinations not in approved baselines
- Flag outbound connections to Google Forms URLs that are not user-initiated, particularly in high-sensitivity environments
- Review access logs for ca[.]iq and similar government infrastructure domains for anomalous outbound connections
- Treat unsolicited RAR archives referencing government ministry documents as high-risk delivery vectors, particularly in diplomatic and foreign ministry environments
- Implement memory-based detection for in-memory PowerShell execution patterns consistent with GHOSTFORM's architecture

9. References

- Zscaler ThreatLabz. (March 2026). Dust Specter APT Targets Government Officials in Iraq. Zscaler.com
- Reuters. (February 28, 2026). US and Israel strike Iran; Supreme Leader Khamenei killed.
- AP News. (November 2025). US envoy warns Iraq of Israeli strikes on Iran-aligned militias.
- Open source geopolitical reporting, 2025-2026 (multiple sources cited throughout).